

■ 1.5.5 Equations

Section 1.2.2 discussed *assignments* such as $x = y$ which *set* x equal to y . This section discusses *equations*, which *test* equality. $x == y$ *tests* whether x is equal to y .

This *tests* whether $2 + 2$ and 4 are equal. `In[1]:= 2 + 2 == 4`
 The result is the symbol `True`. `Out[1]= True`

It is very important that you do not confuse $x = y$ with $x == y$. $x = y$ is an *imperative* statement that actually causes an assignment to be done. $x == y$ merely *tests* whether x and y are equal, and causes no explicit action. If you have used the C programming language, you will recognize that the notation for assignment and testing in *Mathematica* is just the same as in C.

$x = y$	assigns x to have value y
$x == y$	tests whether x and y are equal

Assignments and tests.

This *assigns* x to have value 4 . `In[2]:= x = 4`
`Out[2]= 4`

If you ask for x , you now get 4 . `In[3]:= x`
`Out[3]= 4`

This *tests* whether x is equal to 4 . In this case, it is. `In[4]:= x == 4`
`Out[4]= True`

x is equal to 4 , not 6 . `In[5]:= x == 6`
`Out[5]= False`

This removes the value assigned to x . `In[6]:= x = .`

The tests we have used so far just involve numbers, and always give a definite answer, either `True` or `False`. You can also do tests on symbolic expressions.

Mathematica cannot get a definite result for this test unless you give x a specific numerical value. `In[7]:= x == 5`
`Out[7]= x == 5`

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If you replace x by the specific numerical value 4, the test gives `False`.

```
In[8]:= % /. x -> 4
Out[8]= False
```

Even when you do tests on symbolic expressions, there are some cases where you can get definite results. An important one is when you test the equality of two expressions that are *identical*. Whatever the numerical values of the variables in these expressions may be, *Mathematica* knows that the expressions must always be equal.

The two expressions are *identical*, so the result is `True`, whatever the value of x may be.

```
In[9]:= 2 x + x^2 == 2 x + x^2
Out[9]= True
```

Mathematica does not try to tell whether these expressions are equal. In this case, using `Expand` would make them have the same form.

```
In[10]:= 2 x + x^2 == x (2 + x)
Out[10]= 2 x + x^2 == x (2 + x)
```

Expressions like $x == 4$ represent *equations* in *Mathematica*. There are many functions in *Mathematica* for manipulating and solving equations.

This is an *equation* in *Mathematica*. Subsection 1.5.7 will discuss how to solve it for x .

```
In[11]:= x^2 + 2 x - 7 == 0
Out[11]= -7 + 2 x + x^2 == 0
```

You can assign a name to the equation.

```
In[12]:= eqn = %
Out[12]= -7 + 2 x + x^2 == 0
```

If you ask for `eqn`, you now get the equation.

```
In[13]:= eqn
Out[13]= -7 + 2 x + x^2 == 0
```